

Seifeddine Ben Tekaya - Microbial Ecologist | Environmental Genomics | Quantitative Analysis of Soil-Associated Systems

Laramie, WY, USA

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Professional Summary

Microbial ecologist and research scientist with experience spanning environmental microbiology, microbial genomics, transcriptomics, and quantitative analysis of non-model biological systems. Developed qPCR- and sequencing-based approaches to study *Frankia* population dynamics across soil and rhizosphere systems, linking microbial patterns to environmental context. Isolated and characterized 19 novel extremotolerant bacterial strains from Wyoming desert environments through integrated cultivation, phenotypic characterization, genome sequencing, and comparative genomics. Extensive experience building reproducible workflows for large biological datasets and translating complex measurements into biologically meaningful interpretation through interdisciplinary collaboration.

Core Technical Skills

Environmental and Soil-Linked Research

Soil and rhizosphere microbial ecology, environmental sample processing, microbial isolation and culturing, phenotypic characterization of non-model microbes, microbial adaptation to harsh environments, plant-associated and rock-associated microbiology.

Quantitative and Computational Analysis

Data science, analytics, statistical analysis, differential expression analysis, comparative genomics, metagenomics, transcriptomics, data cleaning and quality control, reproducible workflows, multivariate biological data interpretation, HPC-based pipeline execution in SLURM environments.

Molecular and Sequencing Approaches

qPCR-based analysis, microbiome sequencing, genome sequencing workflows, comparative genomic analysis, RNA-seq analysis from quality control through differential expression.

Programming and Bioinformatics Tools

R (advanced), Bash (advanced), Python (working familiarity), FastQC, read trimming tools, splice-aware aligners, HTSeq-style quantification, DESeq2, Git-based workflows.

Research Experience

Research Scientist

University of Wyoming. Molecular Biology Department | Sept 2023 – June 2025

- Led quantitative analysis of large-scale transcriptomic datasets examining organismal responses to extreme environmental stressors, including spaceflight, radiation, and desiccation.

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- Designed and executed end-to-end RNA-seq workflows covering quality control, preprocessing, alignment, quantification, normalization, and differential expression analysis using reproducible pipelines in R and Bash.
- Integrated complex biological datasets to identify interpretable response patterns across treatments and support mechanistic, environmentally grounded interpretation.
- Collaborated across interdisciplinary teams to connect computational results with experimental design, biological context, and manuscript development.

Postdoctoral Research Associate

University of Wyoming. Department of Botany | Aug 2018 – Aug 2023

- Conducted independent research on environmentally derived microbial systems using genomics, metagenomics, cultivation, and comparative analysis.
- Built and curated a culture collection of extremotolerant microorganisms from arid Wyoming environments and linked genomic features to stress-response phenotypes.
- Isolated and characterized 19 novel microbial species from Wyoming desert and rock-associated systems using cultivation, physiological characterization, genome sequencing, and comparative genomics.
- Developed computational workflows for sequencing-based analysis of non-model environmental organisms and community-level datasets.
- Mentored undergraduate researchers in data analysis, scripting, and biological interpretation.

Selected Research Highlights

- Developed qPCR- and sequencing-based approaches to study ***Frankia* population dynamics across soil and rhizosphere systems**.
- Investigated environmentally derived microbes from **arid Wyoming soils and lithic habitats** through cultivation, genomics, and physiological characterization.
- Built reproducible computational workflows for high-dimensional sequencing datasets from non-model organisms.
- Interpreted biological responses across complex environmental conditions through integrative statistical and comparative analyses.

Selected Research Outputs

Publications

Seifeddine Ben Tekaya, Imen Nouioui, Gabryelle May Flores, Meina Neumann-Schaal, Felix Bredoire, Franco Basile, Linda TA van Diepen, Naomi L Ward (2024)

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Geodermatophilus maliterrae sp. nov., a member of the Geodermatophilaceae isolated from badland surfaces in the Red Desert, Wyoming, USA. International Journal of Systematic and Evolutionary Microbiology. DOI:10.1099/ijsem.0.006603

Seifeddine Ben Tekaya, Trina Guerra, David Rodriguez, Jeffrey O. Dawson, Dittmar Hahn (2018) *Frankia* diversity in host-plant root nodules is independent of abundance or relative diversity of *Frankia* in corresponding rhizosphere soils. Applied and Environmental Microbiology. DOI:10.1128/AEM.02248-17

Seifeddine Ben Tekaya, Abirama Sundari Ganesan, Trina Guerra, Jeffrey Dawson, Michael RJ Forstner and Dittmar Hahn (2017) SybrGreen and Taqman-based q PCR approaches allow assessment of the abundance and relative distribution of *Frankia* clusters in soil. Applied and Environmental Microbiology. DOI: 10.1128/AEM.02833-16

Atef Jaouani, Mohamed Neifar, Valeria Prigione, Amani Ayari, Imed Sbissi, Sonia Ben Amor, **Seifeddine Ben Tekaya**, Giovanna Cristina Varese, Ameer Cherif and Maher Gtari (2014) Diversity and Enzymatic Profiling of Halotolerant Micromycetes from Sebkha El Melah, a Saharan Salt Flat in Southern Tunisia. BioMed Research International. DOI: 10.1155/2014/439197

Seifeddine Ben Tekaya, Sherlyn Tipayno, Murugesan Chandrasekaran, Woojong Yim and Tongmin Sa (2012). Actinobacteria isolation from metal contaminated soils for assessment of their metal resistance and plant growth promoting (PGP) characteristics. Korean Journal of Soil Science and Fertilizers. DOI: 10.7745/KJSSF.2012.45.4.593

Seifeddine Ben Tekaya, Sherlyn Tipayno, Kiyeon Kim, Parthiban Subramanian and Tongmin Sa (2013). Rhizobacteria: Restoration of Heavy Metal Contaminated Soils. In: Physiological Mechanisms And Adaptation Strategies in Plants Under Changing Environment. DOI 10.1007/978-1-4614-8600-8

Ongoing Research

- Spaceflight transcriptomics manuscript integrating RNA-seq differential expression analysis, DESeq2 interaction modeling, functional enrichment, and temporal expression profiling to characterize 7-day and 61-day responses in *Hypsibius exemplaris* and compare them with terrestrial stress responses. Complementary functional testing of six selected genes in HEK cells under simulated microgravity and UV-C exposure is supported by fluorescence microscopy and Western blot confirmation of expression.
- Large-scale microbial genomics program involving the isolation and genomic characterization of nineteen novel microbial species, with multiple species descriptions in advanced preparation.

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- Co-first-author manuscript, “Delayed Proteomic Response to UV Irradiation in *Geodermatophilus obscurus* G20,” investigating temporal changes in the proteome following UV-C exposure.

Education

PhD, Aquatic Resources

Texas State University, USA | 2018

MS, Microbial Ecology

University of Tunis El Manar, Tunisia | 2009

Additional Experience

- Extensive interdisciplinary collaboration across biology, ecology, and computational research teams.
- Experience mentoring students and junior researchers in data-driven biological research.